

ORF309/MAT380

Poisson Processes IV:

Nonhomogeneous Poisson Processes

(pp. 110-113)

Ioannis Akrotirianakis

Agenda for today

- Examples (continued from last lecture)
 - Kidney arrivals
- Nonhomogeneous Poisson Processes

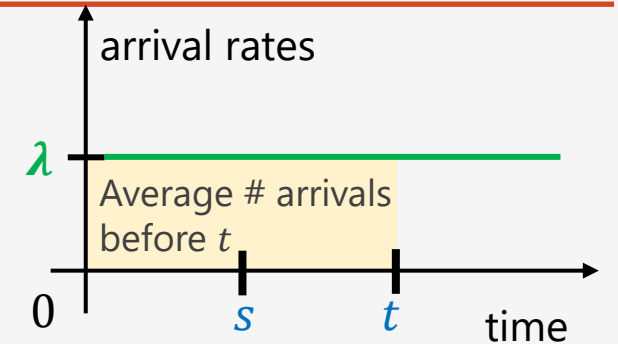
PSET 5: due today @ 11:59 PM

Motivation for varying arrival rates

- Let's recall the **modeling assumptions of Poisson processes**

- For a Poisson process $\{N_t\}_{t \geq 0}$ with rate λ :

- The number of arrivals, $N_t - N_s$, in $(s, t]$ depends on ***length*** $([s, t])$
- The average number of arrivals per time unit is **constant** $\rightarrow \lambda$

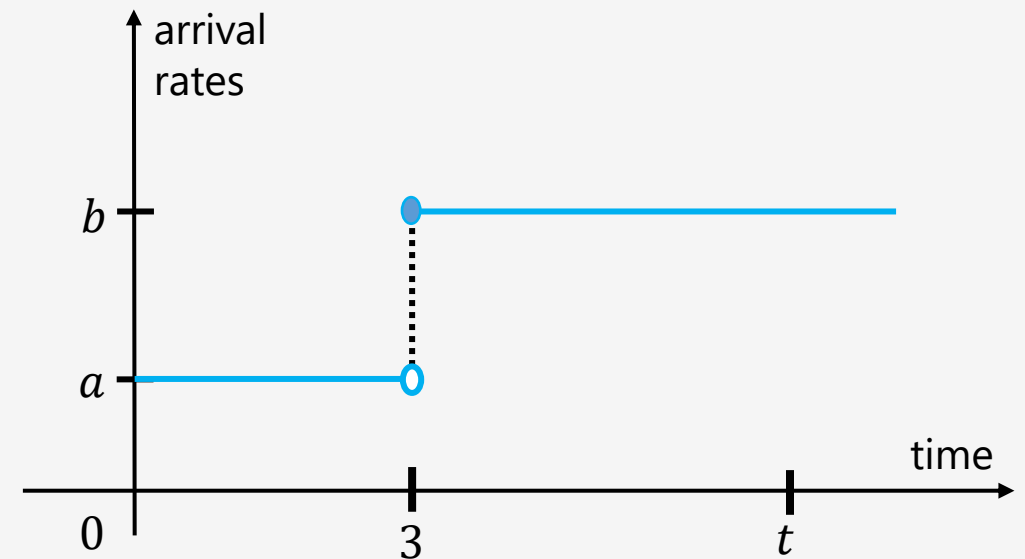


- Realistic for:** emissions of ***photons*** from a laser or ***positrons*** from a PET scanner
- Not realistic:** for the ice cream store in our neighborhood
 - The average # customers **changes** through time $\rightarrow \lambda(t)$ (*function of time*)

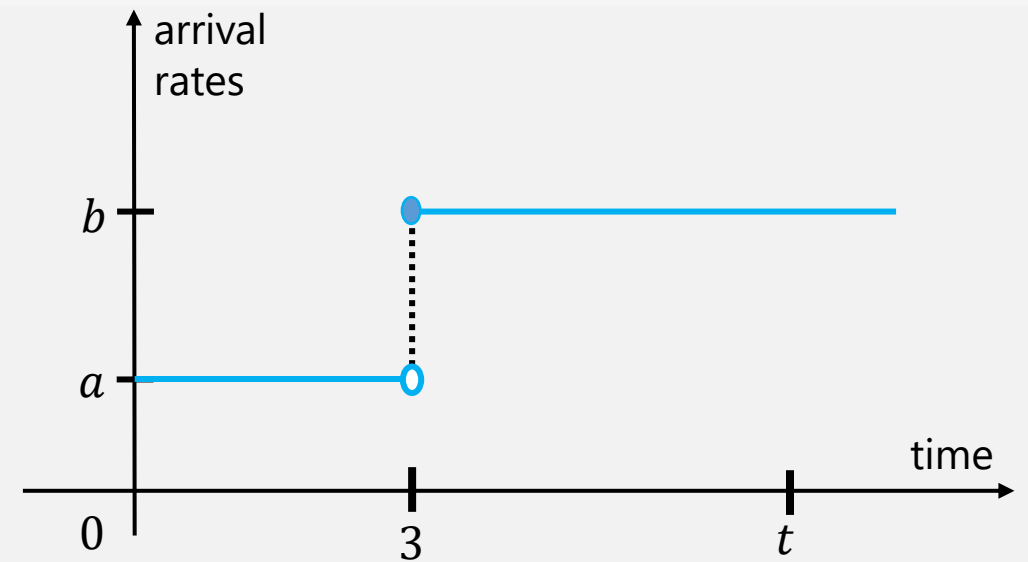
Two different arrival rates

- **Example:**

- Consider an arrival model with **2 different rates**
- **From 0am to 2:59:59...am** customer arrivals follow Poisson with rate a ,
- **After 3:00am** customer arrivals follow Poisson with rate b
- **Question:** what is the distribution of customers arriving by time t , where $t > 3$



- Define: $N_t = \text{"# customers by time } t\text{"}$
- **The challenge**: how can we write the number of customers by $t > 3$??

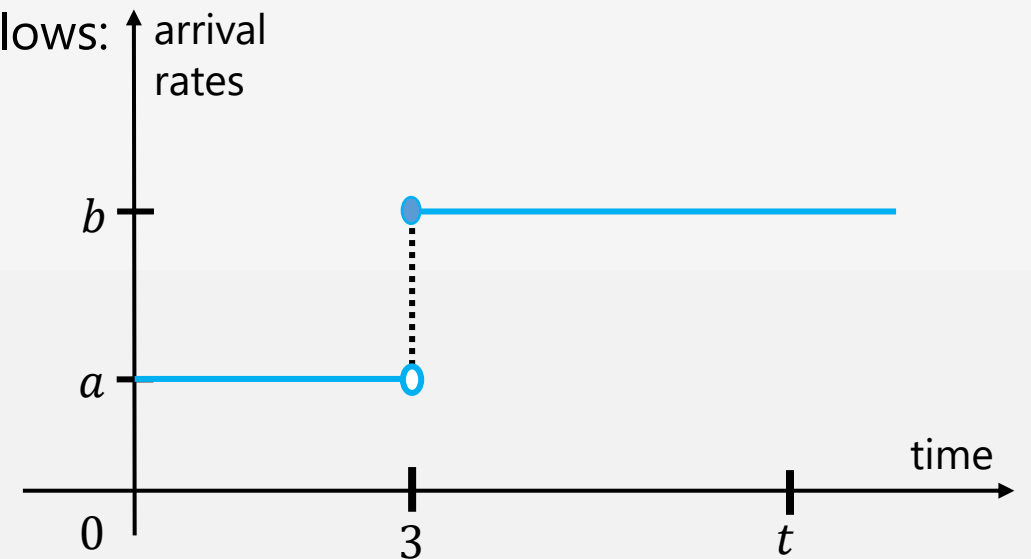


- Define: $N_t = \text{"# customers by time } t\text{"}$

- We can write the number of customers by $t > 3$ as follows:

$$N_t = N_3 + (N_t - N_3)$$

Do we know the distributions of N_3 and $N_t - N_3$



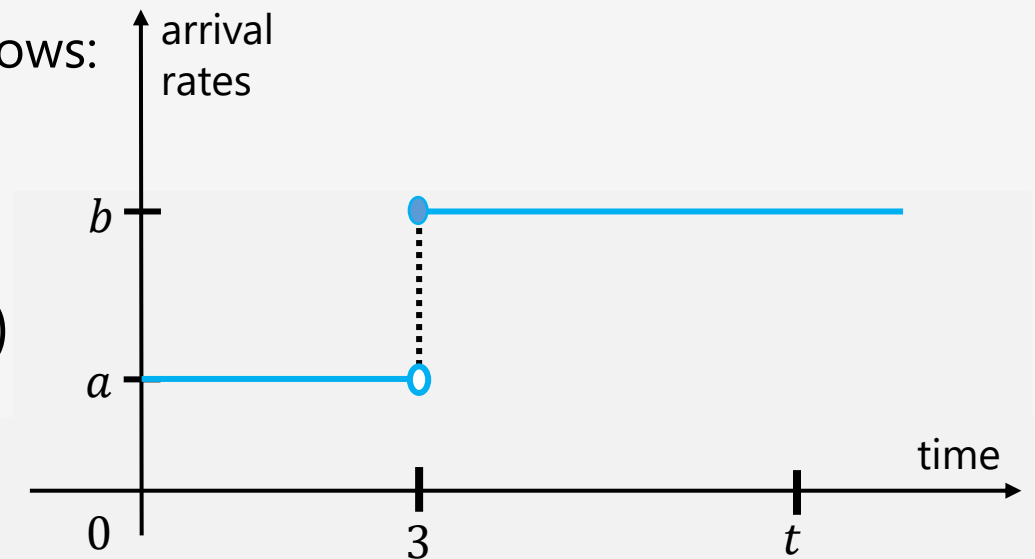
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- YES, we know: $N_3 \sim \text{Pois}(3a)$, $N_t - N_3 \sim \text{Pois}((t - 3)b)$

Are N_3 and $N_t - N_3$ independent??



- Define: $N_t = \text{"# customers by time } t\text{"}$

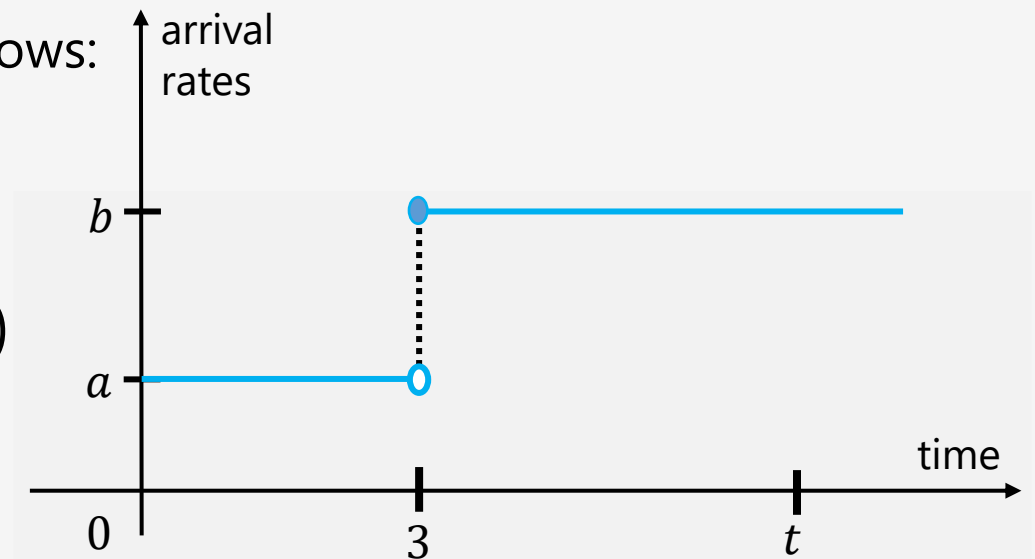
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- Customers arrive independently at each time

- $N_3 \perp\!\!\!\perp N_t - N_3$ (very important!)

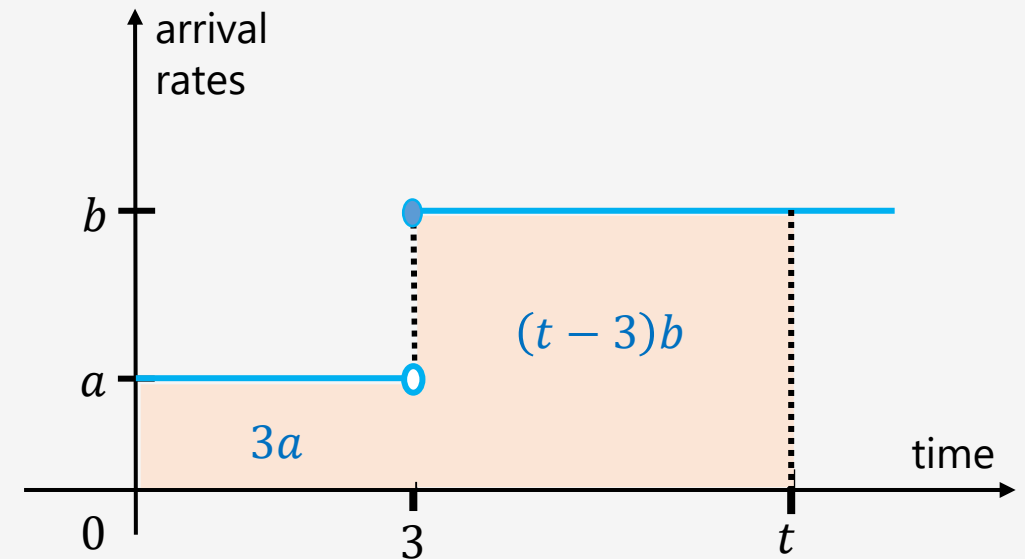


What is the distribution of N_t ??

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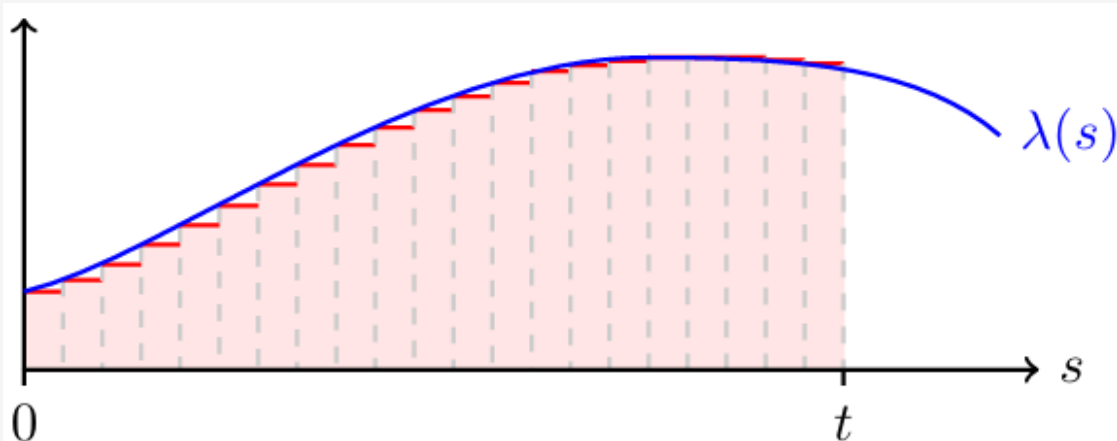
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- Customers arrive independently at each time
 - $N_3 \perp\!\!\!\perp N_t - N_3$ (very important!)
- Hence: $N_t \sim \text{Pois}(3a + (t - 3)b)$



rate = area under rate function

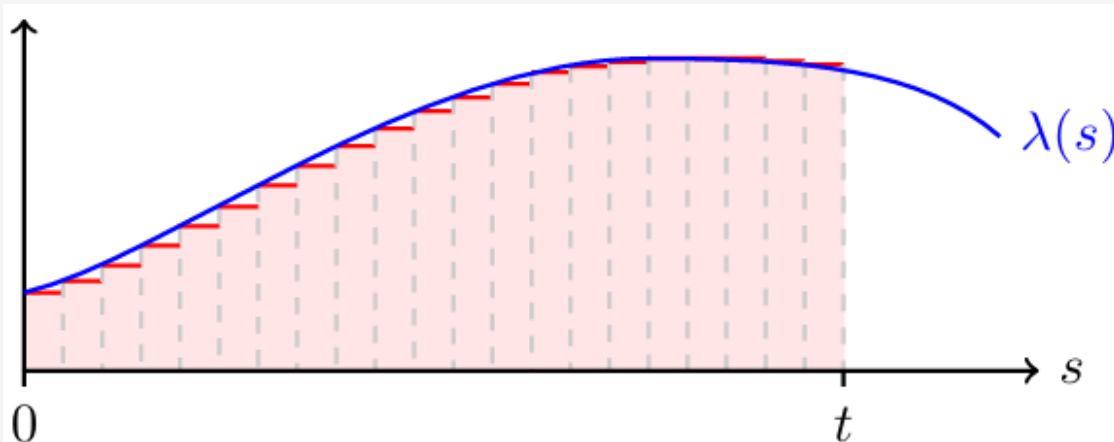
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- We can **generalize** this concept
 - **Arrival rate** can be **any nonnegative function** $\lambda(t) \geq 0$
 - The **number of customers arriving by time t** is Poisson distribution with **mean** equal to the **area under the rate function** $\Lambda(t) = \int_0^t \lambda(s) ds$



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$\lambda(t)$ can be thought of as the **instantaneous rate of arrival occurrence at time t**

In essence, the rate function specifies how often arrivals occur at time t

Nonhomogeneous Poisson processes

- **Definition:** A counting process $\{N_t\}_{t \geq 0}$ is called **nonhomogeneous Poisson process with rate function** $\lambda: R_+ \rightarrow R_+$ if it satisfies the following properties

1. $N_t - N_s$ and $\{N_r\}_{r \leq s}$ are independent for all $t > s$

2. $N_t - N_s \sim \text{Pois} \left(\int_s^t \lambda(u) du \right)$ for all $t > s$

- **Remark:** the increments are still independent (i.e., arrivals occur independently)
- **Remark:** **but the increments are no longer stationary**

You **cannot** say $N_t - N_s \sim N_{t-s} \forall t > s$. Restarting the clock, does **not** work here
- **Remark:** Many of the properties of the standard Poisson processes can be **generalized** to nonhomogeneous Poisson processes

The distribution of N_t in nonhomogeneous Poisson Processes

- The **mean rate** of a nonhomogeneous Poisson process $\{N_t\}_{t \geq 0}$ is defined as

$$\Lambda(t) = \int_0^t \lambda(u) du$$

- The **Probability Mass Function** of N_t :

$$P\{N_t = k\} = \frac{(\Lambda(t))^k \exp(-\Lambda(t))}{k!} = \frac{\left(\int_0^t \lambda(u) du\right)^k \exp\left(-\int_0^t \lambda(u) du\right)}{k!}$$

- The **Cumulative Distribution Function** of N_t

$$P\{N_t \leq k\} = \sum_{i=0}^k \frac{(\Lambda(t))^i \exp(-\Lambda(t))}{i!} = \sum_{i=0}^k \frac{\left(\int_0^t \lambda(u) du\right)^i \exp\left(-\int_0^t \lambda(u) du\right)}{i!}$$

First arrival time & Lifetimes

- **Example:**
 - Consider the nonhomogeneous Poisson process $\{N_t\}_{t \geq 0}$ with rate function $\lambda(t)$.
 - Define T as the **time of the 1st arrival**
 - **Question:** What is the distribution of T ?

Hint: What is the connection between T and N_t ?

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 - **Recall:** for standard Poisson processes we used the equivalence of the events

$$\{T > t\} = \{N_t = 0\} \Rightarrow$$

The 1st arrival is
in the future

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Hence: T is a **Lifetime** with **hazard rate** $\lambda(t)$ and **Hazard function** $\Lambda(t) = \int_0^t \lambda(u) du$

- **Example (cont.):**

- **Remark #1:** you will wonder why we go straight to compute $P\{T > t\}$ instead of $P\{T \leq t\}$ which will give us the CDF of T (the first arrival)?
- The answer is **mathematical convenience** since $P\{T \leq t\}$ is harder to compute directly – it involves infinite sums which although doable, they are cumbersome...

$$P\{T \leq t\} = P\{N_t \geq 1\} = \frac{\sum_{k=1}^{\infty} \exp\left(-\int_0^t \lambda(u) ds\right) \left(\int_0^t \lambda(u) ds\right)^k}{k!}$$

Good luck computing this!

- **Remark #1 (cont.):**

- On the other hand, considering the equivalence of the events

$$\{T > t\} = \{N_t = 0\} \Rightarrow P\{T > t\} = P\{N_t = 0\}$$

we are able to come up with an expression that involves only one term and can be computed easily and accurately without too much pain...:

$$P\{T > t\} = P\{N_t = 0\} = \frac{\exp\left(-\int_0^t \lambda(u) du\right) \left(\int_0^t \lambda(u) du\right)^0}{0!} = \exp\left(-\int_0^t \lambda(u) du\right)$$

-
- **Remark #2:** Considering the event $\{T > t\}$ is easier to come up with, if we think of T as the time until the first failure occurs.
 - This is very useful in Reliability theory, where the 1st failure may result in a machine becoming unavailable (which may also delay production, etc).
 - Then it is easier to think of the event $\{T > t\}$ which means that the 1st failure has not occurred yet, and the system is still functioning normally.
 - Also, for reliability engineers T (i.e., the time of the first failure) is associated with the lifetime of the system, since a failure may knock down the system...
 - The rate of the exponential distribution can be thought of as the hazard function.

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- **Some general applications**

Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

Exercise

- ORFE's print room has 2 printers, A and B, which function independently.
- Jobs done on printer A require X minutes to print, and $X \sim \text{Expon}(\alpha = 2 \text{ jobs/minute})$
- Jobs done on printer B require Y minutes to print, and $Y \sim \text{Expon}(\beta = 3 \text{ jobs/minute})$
- You walk into the room and see both printers are busy.
- **Question 1:** what is the probability the printer B will finish its current job before printer A?
- **Question 2:** what is the probability that at least one printer will finish its current job in the next 30 seconds?
- **Question 3:** what is the probability that both printers will finish their current jobs in the next 30 seconds?

High-Frequency Trading Volume

- The **rate of trades** on a stock exchange is not constant during the day:
 - **High:** soon after the market opens (9:30am) – traders react to last night's news and pre-market activity → we observe a burst of transactions
 - **Low:** during lunchtime (12-1pm); *even traders need to eat...*
 - **High:** after lunch and until market close (4pm) – traders adjust portfolio, or participate in last-minute speculation (close to 4pm).

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- How could you model these trades with the different rates:
 - N_t = #trades in a given time interval $(0, t]$
 - $\lambda(t)$: the **rate function** could follow a **U-shape** with highs in the opening and closing of the market

$$\lambda(t) = \gamma_1 e^{\gamma_2 t} + \gamma_3 e^{\gamma_4 t} + \gamma_5$$

Where t is the elapsed time from the opening of the market, and γ_i are positive constants that we derive by **fitting the model to historical data** (see for example, ORF307 or ORF363)

- The cumulative rate $\Lambda(t) = \int_0^t \lambda(s) ds$ will give the **expected total number of trades up to time t**

Customer arrivals at a casino

- The rate at which new customers arrive at **casino's gambling tables**, or the rate at which **customers place their first bet** on online platforms, depends on day and time of the week.
 - **Weekdays daytime**: the arrival rate is low because people are working
 - **Weekdays (evenings/nights)**: the rate sharply increases as people finished working and perhaps look for short entertainment.
 - **Weekends/Holidays**: The arrival rates are significantly higher throughout the day, compared to weekdays with possible peaks in the evenings

Customer arrivals at a casino

- How could you model these arrivals with the different rates:
 - N_t = # of players arriving at a new blackjack table in the time interval $(0, t]$
 - $\lambda(t)$: the rate function is a periodic function that captures the daily cycle
 - You could model it using **trigonometric functions** of the form

$$\lambda(t) = \lambda_0 + \sum_{k=1}^n \left(a_k \cos\left(\frac{2\pi kt}{24}\right) + b_k \sin\left(\frac{2\pi kt}{24}\right) \right)$$

Where t is time in hours, λ_0 is the **baseline** rate, and a_k, b_k try to capture the **daily peaks**
